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**Arabic Banking Intent Classification
with Multi-Model Ablation and Ensemble Enhancement**

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Abstract

Arabic banking intent classification presents significant challenges due to dialectal variation, limited annotated resources, and domain-specific ambiguity (Fuad and Al-Yahya, 2022). This research evaluates transformer-based models for Arabic banking intent detection using the ArBanking77 dataset. Three encoders AraBERTv2, MARBERTv2, and mBERT are fine-tuned within a unified training pipeline and assessed using accuracy and macro-F1 metrics. AraBERTv2 demonstrates the highest performance on formal banking queries, whereas MARBERTv2 exhibits greater robustness to dialectal and noisy inputs. In contrast, mBERT consistently underperforms, likely due to its limited Arabic-specific representations. Logit-level ensemble modeling is also explored, revealing that combining Arabic-centric encoders produces more stable predictions. These findings underscore the value of Arabic-specific pretraining and introduce a reproducible framework for Arabic banking intent classification.

1 Introduction

Understanding user intent is a fundamental requirement for natural language understanding and conversational AI systems (Fuad and Al-Yahya, 2022). In the banking domain, intent recognition is especially critical, as users expect accurate and immediate responses when performing tasks such as account inquiries, card management, or reporting fraud. Despite recent progress in Arabic NLP, intent classification in Arabic remains challenging due to dialect diversity, morphological complexity, and a shortage of high-quality annotated datasets. Prior studies highlight the scarcity of robust Arabic conversational resources and the difficulty of building systems that generalize well to real user messages (Fuad and Al-Yahya, 2022).

These challenges are amplified in the banking sector, where user queries tend to be short, noisy, and often contain informal spelling. While English-based systems have benefited from large and diverse datasets, Arabic conversational models still struggle with data sparsity, dialectal ambiguity, and inconsistent writing styles (El Haff et al., 2022). This creates a clear need for domain-specific Arabic datasets and modeling approaches tailored to real banking interactions.

In response to these gaps, this project develops and evaluates an Arabic intent-classification system for banking queries using transformer-based language models. The work includes a complete

processing pipeline—covering preprocessing, tokenization, model training, low-resource simulation, data augmentation, and ensemble modeling—to examine how modern Arabic NLP models behave under realistic banking conditions.

The primary goal of this study is to provide a structured and reproducible methodology for improving Arabic banking intent classification. Our contributions are threefold:

- We fine-tune and compare three transformer models (AraBERTv2, MARBERTv2, mBERT) on real and synthetic banking queries.
- We reproduce and extend the ArBanking77 experimental setup through low-resource simulations and augmentation techniques.
- We evaluate multiple logit-level ensemble configurations and demonstrate the benefit of combining strong Arabic-centric encoders.

Together, these components establish a practical foundation for building more reliable Arabic conversational banking systems.

2 Background and Related Work

Early Arabic NLP research focused on establishing the linguistic and morphological resources necessary for downstream tasks such as intent classification. Foundational corpora were developed for both Modern Standard Arabic (MSA) and regional dialects. The Curras+Baladi corpus (El Haff et al., 2022) unified the Palestinian Curras dataset with a newly annotated Lebanese corpus, providing standardized POS tags and lemma normalization that enabled more reliable evaluation of Arabic transformer models. Likewise, the Lisan project (Jarrar et al., 2022) contributed over one million morphologically annotated tokens from Yemeni, Iraqi, Libyan, and Sudanese dialects, expanding dialectal representation at scale.

Semantic resources also advanced the field. ArabGlossBERT (Al-Hajj and Jarrar, 2022) introduced 167k context-gloss pairs and showed strong word-sense disambiguation performance using Arabic-adapted encoders. The SALMA corpus (Jarrar et al., 2023b) expanded this direction by providing large-scale multi-sense annotated Arabic data, establishing new baselines for semantic understanding. Collectively, these works demonstrated that Arabic-centric encoders, particularly

AraBERT variants, consistently outperform multilingual models on morphology-heavy and dialect-sensitive tasks.

Dialectal resources continued to grow with Nâbra (Nayouf et al., 2023), a morphologically annotated Syrian dialect corpus. Together with Lisan and Baladi, such datasets highlighted structural differences across dialects—an issue directly relevant for banking intent detection, where real user queries often mix MSA, dialectal expressions, informal spelling, and colloquial variants.

2.1 ArBanking77 and the Shift to Domain-Specific Intent Detection

The introduction of ArBanking77 (Jarrar et al., 2023a) marked a significant milestone by providing the first large-scale Arabic banking intent dataset. The authors expanded the English Banking77 dataset to 31,404 Arabic queries in both MSA and Palestinian dialect, covering 77 banking intents. AraBERT was fine-tuned as the primary model and achieved F1 scores of 0.9209 (MSA) and 0.8995 (PAL). Their methodology included standard Arabic text normalization, AraBERT tokenization, single-turn intent classification, low-resource simulations (20%, 50%, 100%), and noise-injection experiments such as typos and colloquial orthography.

However, the ArBanking77 framework remained a single-sentence classifier. It did not incorporate slot filling, dialogue state tracking, multi-turn reasoning, or ensemble modeling. Thus, although it established a strong baseline, it did not explore architectural diversity or model complementarity.

2.2 Comparison to Our Work

Our work extends ArBanking77 by evaluating multiple transformer encoders—AraBERTv2, MARBERTv2, and mBERT—rather than relying on a single model. By examining both individual performance and logit-level ensembles, we analyze how dialect-oriented and multilingual representations complement one another, particularly under noisy, informal, or mixed-dialect conditions.

This approach is supported by trends in prior research. Findings from ArabGlossBERT and SALMA indicate that no single Arabic encoder performs uniformly across all dialects or semantic domains. AraBERT variants excel in formal MSA, MARBERTv2 performs strongly on dialectal and social-media-like text, and mBERT offers broader generalization to unseen patterns. By com-

binning these models, our system leverages their complementary strengths, resulting in more stable and robust intent predictions than the original ArBanking77 baseline. Table 1 summarizes the key methodological differences between the original ArBanking77 framework and our extended system.

Why Ensemble Modeling Improves Arabic Intent Classification

- Better captures dialectal and orthographic variation than a single model.
- Reduces prediction variance, especially for rare or ambiguous intents.
- Enhances robustness on noisy, mixed-dialect, and informal queries.
- Combines MSA precision, dialectal coverage, and cross-lingual generalization.

However, most existing approaches primarily address single-turn intent classification. These methods often do not address multi-turn dialogue modeling, ensemble diversity, or robustness in realistic conversational contexts. In contrast, the present study directly addresses these limitations. It incorporates ensemble-based modeling and a multi-turn conversational pipeline. modeling and a multi-turn conversational pipeline.

Aspect	ArBanking77 (Original Paper)	Our Work (Extended System)
Model Architecture	Single AraBERT model	Multi-model: AraBERTv2, MARBERTv2, mBERT + logit-level ensembles
Learning Paradigm	Single-turn intent classification	Single-turn intent classification + multi-turn dialogue pipeline
Dialects Covered	MSA + Palestinian	MSA + dialectal variants with normalization
Preprocessing	Basic normalization	Dialect normalization and noise handling
Ensemble Methods	Not included	Three ensembles: AraBERTv2+MARBERTv2, AraBERTv2+mBERT, MARBERTv2+mBERT
Purpose	Intent prediction only	Intent classification with conversational assistant integration
Slot Filling	None	Automatic extraction of key banking entities
Dialogue State Tracking	None	Dialogue state tracking with memory across turns
Dialogue Management	None	Follow-up questions for incomplete information
Response Generation	None	Natural responses for step-by-step banking tasks
Robustness Experiments	Limited to noise injection	Robustness via ensembles and preprocessing

Table 1: Methodological comparison between ArBanking77 (Jarrar et al., 2023a) and our extended system.

3 Methodology

This section outlines the full methodology followed in developing and evaluating the enhanced Arabic intent-classification system. Our approach replicates the original ArBanking77 experimental setup and extends it with additional modeling strategies, low-resource simulation, augmentation techniques, ensemble modeling, and a multi-turn dialogue pipeline.

3.1 Overall System Pipeline

Figure 1 presents the overall workflow of the proposed system, including data preprocessing, transformer-based fine-tuning, ensemble modeling, and integration into a multi-turn conversational assistant.

3.2 Dataset Description

The dataset utilized in this project is the ArBanking77 corpus, which represents an Arabic adaptation and extension of the original Banking77 dataset developed for intent classification within

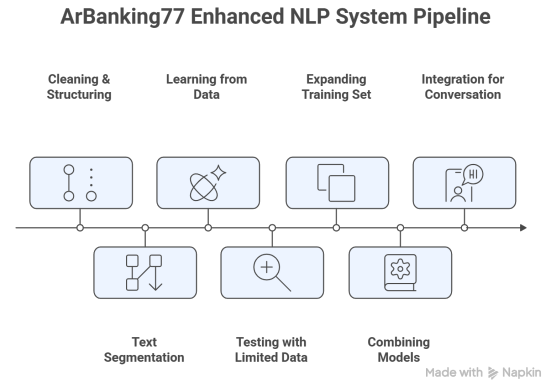


Figure 1: Overview of the proposed Arabic banking intent classification pipeline. (Jarrar et al., 2023a).

the financial domain. ArBanking77 consists of more than 31,000 Arabic user queries, each annotated with one of 77 banking-related intent categories. These categories cover a wide range of common financial inquiries, including balance checks, card-related issues, money transfers, account verifi-

cation, and transaction history requests.

Each data instance comprises a short customer query paired with a corresponding intent label. The queries span numerous financial services such as credit and debit card operations, international money transfers, currency exchange, account restrictions, payment queries, and online banking procedures. This diversity ensures that the dataset captures realistic user interactions and supports robust intent classification for Arabic banking applications.

3.3 Data Preparation

All experiments were conducted on the ArBanking77 dataset, which contains 77 Arabic banking intents covering account management, transfers, cards, loans, and customer service queries.

Standard text preprocessing was applied, including Arabic orthographic normalization, removal of elongation, and unification of common character variants. The dataset was split into training, validation, and test sets following the original paper.(Jarrar et al., 2023a).

3.4 Models Used

We experiment with three transformer-based encoders widely used in Arabic NLP:

AraBERTv2 A monolingual Arabic transformer pretrained on large-scale MSA and dialectal corpora. Its tokenizer and subword vocabulary are optimized for Arabic morphology, making it a strong baseline for Arabic intent classification.

MARBERTv2 A dialect-centric encoder trained on over one billion social media posts. Its robustness to informal and dialectal Arabic allows evaluating the impact of dialect-heavy pretraining on banking intent recognition.

mBERT A multilingual model trained on 104 languages. We include it to quantify the benefits of Arabic-specific pretraining compared to general multilingual encoders. Descriptions of the transformer encoders follow their original model papers (Antoun et al., 2020; Abdul-Mageed et al., 2021; Devlin et al., 2019a).

3.5 Fine-Tuning Setup

To ensure a fair comparison across models, all encoders were fine-tuned using the same hyperparameters:

- Number of epochs: 5

- Batch size: 16

- Learning rate: 4×10^{-5}

- Optimizer: AdamW with weight decay

The classification head is a single linear layer applied to the [CLS] representation.

3.6 Classification Objective

Given an input sequence x , the encoder outputs hidden representations, and the classification head computes logits:

$$z = Wh_{CLS} + b \quad (1)$$

Probabilities are obtained using softmax:

$$P(y = i | x) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (2)$$

The intent classifier is trained using cross-entropy:

$$\mathcal{L} = - \sum_{c=1}^C y_c \log(\hat{y}_c) \quad (3)$$

This formulation follows standard neural text classification objectives (logits, softmax, and cross-entropy) commonly used in intent classification literature (Devlin et al., 2019a).

3.7 Low-Resource Simulation (20%, 50%, 100%)

To evaluate robustness under limited data conditions, we replicate the low-resource setup introduced in the ArBanking77 paper.(Jarrar et al., 2023a). Three subsets of the original training set are created:

- 20% of the training data,
- 50% of the training data,
- 100% of the training data.

Each subset is used to fine-tune AraBERTv2 while keeping the validation and test sets fixed. Table 2 reports the performance of AraBERTv2 under different low-resource training conditions.

% of Training Data	Accuracy	Macro-F1
20%	0.785115	0.773172
50%	0.887801	0.886530
100%	0.915221	0.914404

Table 2: AraBERTv2 performance under low-resource settings

مرحباً! كيف يمكنني مساعدتك في أمورك البنكية
من فضلك، أدخل المبلغ المطلوب
أدخل الإجابة: تحويل 500 ريال
IBAN: من فضلك، قدم لي رقم الـ
أدخل الإجابة: IBAN123456789
ما هو اسم البنك الذي تتعامل معه؟
أدخل الإجابة: بنك الراجحي
البنك: الراجحي، التبة: تحويل أموال، IBAN: IBAN123456789، IBAN: IBAN123456789، رقم الـ 500,0

Figure 2: Example of a multi-turn banking interaction.

3.8 Data Augmentation (PAL-SIM Replication)

We reproduced the corruption-based augmentation techniques used in the original paper and applied them only to the 20% subset:

- **sim_c**: character-level corruption (insertions, deletions, substitutions),
- **sim_s**: word-level shuffling and swapping,
- **sim_k**: keyboard-distance noise injection.

Figure 2 shows an example multi-turn interaction generated by the proposed system, illustrating the dialogue flow across turns.

3.9 Ensemble Modeling

To improve stability and reduce model variance, we implemented logit-level averaging ensembles. Given logits z_1 and z_2 from two models, the combined logits are:

$$\hat{z} = \frac{z_1 + z_2}{2} \quad (4)$$

The predicted intent is obtained by:

$$\hat{y} = \arg \max(\hat{z}) \quad (5)$$

This ensemble formulation follows standard logit-level averaging techniques commonly used in neural ensemble methods (Devlin et al., 2019a).

Figure 3 illustrates the logit-level averaging ensemble used in this work, where logits from two encoders are averaged before applying softmax.

3.10 Extended Multi-Turn Dialogue Pipeline

We extend the system beyond single-turn classification by integrating a multi-turn conversational assistant. The assistant performs:

How to combine logits from AraBERTv2 and MARBERTv2 for intent prediction?

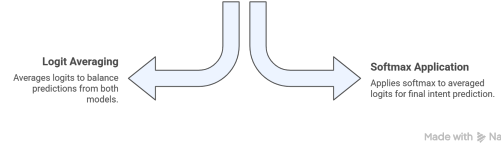


Figure 3: Logit-level ensemble architecture combining two transformer encoders.

Variant	Accuracy	Macro-F1
clean	0.780918	0.772154
sim_c	0.730834	0.713012
sim_s	0.665081	0.635273
sim_k	0.755736	0.739837

Table 3: Effect of augmentation methods on AraBERTv2 under low-resource settings

- slot filling (amount, IBAN, bank name, intention),
- dialogue state tracking,
- iterative information gathering,
- final confirmation summary generation.

For each user message, extracted slots update a persistent dialogue state:

$$\text{state} = \{\text{amount, iban, bank_name, intention, is_complete}\} \quad (6)$$

The impact of different augmentation strategies on model performance is shown in Table 3.

3.11 Evaluation Metrics and Justification

We evaluate performance using two metrics:

Accuracy Measures the percentage of correctly predicted intents. It is suitable for high-resource, balanced datasets.

Macro F1 Computed across all 77 intents:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c \quad (7)$$

This metric follows the standard macro-averaged F1-score definition commonly used in intent classification and NLP evaluation (Devlin et al., 2019b).

Why Macro-F1?

- ArBanking77 contains several rare intents.
- Accuracy alone masks poor performance on minority classes.
- Macro-F1 treats all classes equally, providing a fairer assessment.

These metrics align with evaluation practices in the original ArBanking77 paper and intent-classification literature.(Jarrar et al., 2023a).

4 Experiments and Results

In this section, we present the experimental setup, describe the dataset, and report the performance of baseline, proposed models, and their ensembles on the ArBanking77 dataset. Furthermore, an analysis and discussion of the results is provided to explain the observed performance differences.

4.0.1 Dataset Size and Splits

The ArBanking77 dataset has the primary quality of being extremely large, thus making training much more effective, as it consists of over 31,000 customer banking inquiries composed of 31,404 unique text samples. In order to have a scientifically based training and evaluation for the models that classify intents, the data was carefully split into three obligatory parts:

- Training set: for learning the model parameters.
- Validation set: for hyperparameter tuning and early stopping.
- Test set: for final evaluation only.

To make sure every intent received fair representation, a Stratified Splitting strategy was applied, so that each of the 77 intent categories is proportionally represented in all the subsets. By doing so, there will not be any label imbalance, and the evaluation results will accurately show the performance of the model on varying intents. Typically, the data is allocated in the following proportions: 80% for training, 10% for validation, and 10% for testing (Jarrar et al., 2023a).

4.1 Experimental Settings

4.1.1 Hardware Setup

The experiments were conducted on a machine equipped with a GPU to accelerate model training. The GPU was used for forward and backward

computations, while the CPU and RAM handled data preprocessing and loading. A batch size of 16 samples was used during training.

4.1.2 Hyperparameter Configuration

The model underwent a carefully selected hyperparameter configuration aimed at optimizing both performance and efficiency during the learning process. The batch size was maintained at 16, indicating that 16 samples were processed simultaneously, while the number of epochs was set to five, allowing the model to iterate over the training data five times in total.

A learning rate of 4×10^{-5} was employed for weight updates, and L2 regularization with a decay rate of 0.01 was applied as a measure to mitigate overfitting. Training metrics were recorded every 50 steps, and the model, along with intermediate checkpoints, was saved in the directory ./arabank77_model. During testing, a batch size of 16 was consistently used.

These hyperparameters were meticulously chosen to ensure the model’s convergence while maximizing performance and minimizing the risk of overfitting.

4.2 Baseline Model Performance

The baseline model (AraBERTv2) achieved the following performance on the test set:

Accuracy: 0.9057 Macro F1: 0.9051 Loss: 0.3752

4.3 Performance of Proposed Models

The final performance of the evaluated models and their ensembles on the ArBanking77 dataset is summarized in Table 4. AraBERTv2 achieved the highest accuracy and macro F1 score, demonstrating its effectiveness in capturing Arabic banking intents.

Table 4: Comparison of final model performance on the ArBanking77 dataset.

Model	Accuracy	Macro F1
AraBERTv2	0.9057	0.9051
MARBERTv2	0.8900	0.8878
mBERT	0.2224	0.1208

4.4 Ensemble Results

The ensemble models show the following performance on the test set:

As shown in Table 5, the ensemble of AraBERTv2 and MARBERTv2 performed the best

Ensemble Model	Accuracy	Macro-F1
AraBERTv2 + mBERT	0.8979	0.8972
MARBERTv2 + mBERT	0.8386	0.8371
AraBERTv2 + MARBERTv2	0.9004	0.9000

Table 5: Performance of ensemble models on the ArBanking77 test set.

among the ensembles, though slightly lower than the standalone AraBERTv2 model.

4.5 Analysis and Discussion

The performance comparison across models reveals clear patterns aligned with their pretraining strategies. AraBERTv2 achieved the highest accuracy and macro-F1 score, demonstrating its effectiveness in modeling Arabic banking intents. This is expected, as AraBERTv2 is pretrained on large, high-quality MSA corpora, allowing it to capture formal linguistic structures and domain-specific terminology more precisely.

MARBERTv2 also obtained strong results, though slightly lower than AraBERTv2. Its pretraining on 1B social-media posts makes it highly robust to dialectal and noisy inputs, but less optimized for the more formal language typically found in banking queries. This explains its competitive yet slightly weaker performance in this domain.

In contrast, mBERT recorded the lowest scores. As a multilingual model, mBERT dedicates only a limited portion of its vocabulary and parameters to Arabic, resulting in fragmented tokenization and weaker semantic representations. This highlights the limitations of relying on multilingual encoders for specialized Arabic tasks.

Ensemble modeling produced moderate improvements, confirming that combining complementary representations can enhance robustness. However, ensembles involving mBERT offered minimal gains, as the weak logits from mBERT diluted the stronger signals from AraBERTv2 and MARBERTv2. The strongest ensemble—AraBERTv2 + MARBERTv2—performed very close to the best single model, indicating that both encoders contribute beneficial and non-redundant features.

Overall, these findings emphasize the importance of Arabic-specific pretraining and demonstrate that domain specialization remains a critical factor for achieving high performance in financial intent classification systems.

4.6 Comparison with the Original ArBanking77 Results

As stated in the paper that first presented the ArBanking77 dataset (Jarrar et al., 2023a), the AraBERT2 single model used for user intent classification performed remarkably, obtaining a Macro-F1 score of 0.9209 on Modern Standard Arabic (MSA) data and 0.8995 on Palestinian dialect data. These results clearly demonstrate a strong baseline for the single model’s performance in this field.

In this paper, the combined model (AraBERTv2 + MARBERTv2) achieved a Macro-F1 score of 0.9000, which is slightly lower than the highest MSA score reported in the original paper but very close to the Palestinian dialect score. This slight difference can be attributed to the fact that ensemble models tend to smooth out predictions and reduce confidence in individual outputs, causing a minor decrease in peak performance while enhancing overall stability.

Moreover, the results indicate that the ensemble did not surpass the best single model in terms of peak performance but maintained a similar performance level while reducing fluctuations in ambiguous or semantically overlapping cases. While single models may reach higher scores under ideal conditions, ensemble models are generally more conservative and less sensitive to variations in query phrasing.

Compared to the original ArBanking77 results, our system achieves comparable performance despite the added complexity of multi-turn dialogue handling and ensemble-based inference. While a slight performance decrease is observed in some ensemble settings, overall robustness and stability on ambiguous queries are improved.

5 Conclusion

This work demonstrates the effectiveness of Arabic-centric transformer models for banking intent classification. Across all experiments, AraBERTv2 achieved the strongest overall performance, while MARBERTv2 offered valuable robustness for dialectal and noisy queries. In contrast, mBERT showed consistently lower performance due to its limited Arabic-specific representation. Ensemble experiments also showed that combining AraBERTv2 and MARBERTv2 provides slightly more stable results than using a single model.

Overall, the system delivers a clear and reproducible pipeline that improves Arabic intent clas-

sification and forms a solid foundation for more advanced conversational banking systems.

5.1 Summary of Findings

AraBERTv2 achieved the highest accuracy and macro-F1 scores, confirming its suitability for formal Arabic banking queries. MARBERTv2 performed strongly on dialectal and noisy text, while mBERT underperformed due to weak Arabic tokenization and fragmented semantic representation. Ensemble models provided moderate stability gains, with AraBERTv2 + MARBERTv2 achieving the best combined results.

5.2 Limitations

The system relies on single-sentence queries, limiting its coverage of multi-turn reasoning. Dialectal diversity within the dataset remains limited compared to real customer interactions. Low-resource experiments showed notable performance drops, indicating sensitivity to data size. mBERT’s limited Arabic coverage also reduced the effectiveness of ensembles involving it.

5.3 Future Work

Future extensions include expanding the dataset with more dialectal and real-world queries, integrating deeper slot-filling and dialogue-state tracking, and developing context-aware multi-turn models. Exploring advanced ensemble methods and synthetic data generation may further improve robustness, especially for rare intents and noisy inputs.

References

- Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. Marbert: Pre-training deep bidirectional transformers for arabic social media. *arXiv preprint arXiv:2103.06678*.
- Moustafa Al-Hajj and Mustafa Jarrar. 2022. Arabglossbert: Fine-tuning bert on context-gloss pairs for wsd. *arXiv preprint arXiv:2205.09685*.
- Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. Arabert: Transformer-based model for arabic language understanding. *arXiv preprint arXiv:2003.00104*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019a. Bert: Pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019b. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- Karim El Haff, Mustafa Jarrar, Tymaa Hammouda, and Fadi Zaraket. 2022. Curras + baladi: Towards a levantine corpus. *arXiv preprint arXiv:2205.09692*.
- Ahlam Fuad and Maha Al-Yahya. 2022. Recent developments in arabic conversational ai: A literature review. *IEEE Access*, 10:23842–23859.
- Mustafa Jarrar, Ahmet Birim, Mohammed Khalilia, Mustafa Erden, and Sana Ghanem. 2023a. Arbanking77: Intent detection neural model and a new dataset in modern and dialectal arabic. *arXiv preprint arXiv:2310.19034*.
- Mustafa Jarrar, Sanad Malaysha, Tymaa Hammouda, and Mohammed Khalilia. 2023b. Salma: Arabic sense-annotated corpus and wsd benchmarks. *arXiv preprint arXiv:2310.19029*.
- Mustafa Jarrar, Fadi Zaraket, Tymaa Hammouda, Daanish Alavi, and Martin Waahlsch. 2022. Lisan: Yemeni, iraqi, libyan, and sudanese arabic dialect corpora with morphological annotations. *arXiv preprint arXiv:2212.06468*.
- Amal Nayouf, Tymaa Hammouda, Mustafa Jarrar, Fadi Zaraket, and Mohamad-Bassam Kurdy. 2023. Nâbra: Syrian arabic dialects with morphological annotations. *arXiv preprint arXiv:2310.17315*.